PANKAJ SINGH

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PROFILE SUMMARY

Python & AI Developer with hands-on experience building and deploying machine learning, deep learning, and LLM-powered applications end-to-end — from data pipeline and model development to API deployment. Comfortable working across classical ML, computer vision, and modern NLP/LLM tooling (LangChain, RAG, Transformers), with a strong foundation in writing clean, production-ready Python code. Fast learner with a research-oriented mindset, able to translate business problems into working AI solutions and ship them independently.

CORE STRENGTHS

- Research-Oriented Problem Solving — skilled at breaking down ambiguous problems into structured, testable ML/AI solutions
- Leadership & Ownership — experience independently driving AI projects from data collection to deployed, working product
- Rapid Prototyping — comfortable moving fast from idea to working proof-of-concept using modern AI tooling
- Cross-Domain Adaptability — equally capable across classical ML, computer vision, and LLM/NLP-based systems
- Communication — able to explain model behavior and technical trade-offs clearly to non-technical stakeholders
- Fast Learner — quick to adopt new models, frameworks, and AI platforms as the field evolves

PROJECTS

AI Document Assistant — RAG-Based Q&A Chatbot

Tech Stack: *Python — LangChain — OpenAI API — FAISS — FastAPI — Streamlit*

- Built a Retrieval-Augmented Generation (RAG) system that lets users upload PDFs/docs and ask natural-language questions, with answers grounded strictly in retrieved context to minimize hallucination.
- Built a document ingestion pipeline with recursive chunking (~500-token windows) and embedding generation, indexed in FAISS for sub-second similarity search across thousands of chunks.
- Tuned prompt templates and top-k retrieval settings to improve answer relevance, reducing irrelevant/incorrect responses in manual evaluation on a test question set.
- Exposed the pipeline via a FastAPI backend with a Streamlit chat UI, including inline source citations for every generated answer.

Real-Time Object Detection & Tracking System

Tech Stack: *Python — YOLOv8 — OpenCV — PyTorch — Flask*

- Fine-tuned a YOLOv8 model on a custom-labeled dataset to detect and track multiple object classes in real time from a live webcam or video feed.
- Optimized inference pipeline (batching, frame resizing, confidence thresholding) to sustain real-time frame rates on standard hardware without a dedicated GPU.
- Implemented multi-object tracking with persistent IDs across frames using centroid/IoU-based association, enabling analytics like footfall counting.
- Wrapped the pipeline in a Flask API with a live MJPEG streaming endpoint, making it easy to plug into other applications or dashboards.

Customer Churn Prediction Engine

Tech Stack: *Python — Pandas — scikit-learn — XGBoost — FastAPI*

- Built an end-to-end ML pipeline predicting customer churn from behavioral and transactional data, covering cleaning, feature engineering, training, and deployment.
- Engineered features (usage recency, frequency, support-ticket history) and ran EDA to surface the strongest churn indicators, improving model signal quality.
- Trained and benchmarked Logistic Regression, Random Forest, and XGBoost using cross-validation, selecting the best model based on ROC-AUC and precision/recall trade-offs.
- Deployed the final model behind a FastAPI endpoint returning real-time churn probability scores, paired with a dashboard visualizing at-risk customer segments.

AI Resume Screener & Job-Match Ranking System

Tech Stack: *Python — spaCy — NLTK — scikit-learn — Sentence-Transformers*

- Built an NLP pipeline that parses resumes (PDF/DOCX), extracting structured skills, experience, and education using named entity recognition and rule-based parsing.
- Used sentence-transformer embeddings and cosine similarity to automatically score and rank candidate resumes against a target job description.
- Added a keyword-gap analysis feature that highlights missing skills relative to the job description, giving both recruiters and applicants actionable feedback.
- Packaged the pipeline into a lightweight web interface where uploading a resume and job description instantly returns a match score and skill breakdown.

SKILLS

Programming Languages: Python (primary), JavaScript, SQL

AI/ML Frameworks: TensorFlow, PyTorch, scikit-learn, Keras, XGBoost

NLP & LLMs: LangChain, Hugging Face Transformers, OpenAI API, spaCy, NLTK, RAG, Prompt Engineering

Computer Vision: OpenCV, YOLOv8, MediaPipe

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Jupyter, EDA & Feature Engineering

Backend & Deployment: FastAPI, Flask, Docker, REST APIs, Streamlit

Databases & Vector Stores: MySQL, MongoDB, FAISS, Pinecone, ChromaDB

Tools & Platforms: Git, GitHub, MLflow, Postman, VS Code, Google Colab, AWS/GCP basics

EDUCATION

B.Tech — Computer Science & Engineering **2023 – 2026**

Shivalik College of Engineering, Dehradun, Uttarakhand

Focus on software development, AI, and system design.

Diploma — Computer Science & Engineering **2021 – 2023**

Govt. Polytechnic Kashipur, Uttarakhand

Specialized in web and mobile development, mastering foundational programming concepts.

Schooling **2021**

Pt. G. B. Pant I C, Kashipur, Uttarakhand

Completed schooling with a strong foundation in mathematics and computer science.